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Unraveling the Interaction Between Physiological Signals and Facial Landmarks on Video Brightness

data-to-paper

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Abstract

The integration of physiological signals and facial landmarks in video analytics holds immense promise for enhancing applications in health monitoring and facial recognition. Despite extensive advancements, the interaction effects between photoplethysmography (PPG) signals and facial landmarks on video brightness remain underexplored in existing literature. Addressing this gap, our study investigates the combined influence of these variables on video brightness variations. Utilizing a dataset comprising video brightness metrics, PPG signals, and facial landmark deviations, we applied multiple linear regression models to assess interaction effects. Our findings indicate significant variability in video brightness predictions attributable to individual fluctuations in PPG signals and facial landmarks, yet their interaction does not significantly impact brightness variability. This research highlights the unexplained variance in our model, suggesting the need for deeper exploration into other potential contributing factors. The insights derived from this study could potentially guide the development of more resilient video analytic systems capable of adapting to fluctuating physiological and environmental conditions.

Introduction

The integration of physiological signals and facial landmark data in video analytics is emerging as a crucial frontier in both health monitoring and facial recognition systems [1, 2]. As digital video platforms proliferate, enhancing the quality and reliability of these applications requires a deep understanding of the interactions between various signals that influence the ultimate video output. In particular, photoplethysmography (PPG) signals embedded in video data offer the potential to infer critical physiological parameters [3], while precise facial landmark localization contributes to robust

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facial recognition capabilities [1]. Understanding how these variables interact to influence video characteristics such as brightness can help optimize the development of next-generation video analytic systems.

Despite significant advancements in video quality enhancement through predictive modeling [4, 5], questions remain about the specific interactions between physiological and facial elements. Prior research has explored perceptual video quality assessment using physiological signals [6] and examined coding techniques to accommodate brightness variations [5]. Nonetheless, it remains unclear how these physiological and structural features interact to influence video brightness specifically. Less is known about the potential for synergistic or antagonistic interactions between PPG signal variability and facial landmark deviations in this context.

The current study addresses this gap by investigating the interaction effects of PPG signal variability and facial landmark deviations on video brightness. While previous studies have explored facial-video-based physiological signal measurement [7], this paper specifically focuses on analyzing the interaction of these variables to provide insights into their combined effect on video characteristics. Using the SCAMPS dataset, this study leverages descriptive and inferential statistical methods to elucidate the role of these interactions in contributing to video brightness variability [8].

Our methodological approach entailed the use of multiple linear regression models with interaction terms to explore the relationship between the studied variables [9]. The model incorporated interaction terms to specifically gauge the combined influence of physiological and facial variables [10]. Our analysis revealed substantive individual variabilities within physiological and facial landmark data. However, our findings indicate that the interaction of these variabilities does not significantly drive changes in video brightness, a conclusion underscored by the low R-squared value obtained [11]. These results underscore the complexity of video brightness variability and suggest the need for future research to incorporate additional variables.

Results

First, to understand the variability in our primary measurements, we examined the descriptive statistics for the standard deviations of video brightness, photoplethysmography (PPG) signals, and facial landmarks. As detailed in Table 1, the mean standard deviation for video brightness was 47.41 with a standard deviation of 16.88. The PPG signals exhibited a remarkably small standard deviation of 1.081 10^{-8} around a mean value of 0.9992, in-

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dicating minimal variation. Conversely, facial landmarks exhibited a more substantial standard deviation of 4.56 from a mean of 37.41. These statistics, derived from data encompassing 2768 observations, reveal the inherent variability present in each measured domain, which sets the stage for further analysis into their interaction effects.

Table 1: Descriptive Statistics of Standard Deviations

	mean	std	count	CI Low	CI High
Video Std	47.41	16.88	2768	46.78	48.04
PPG Std	0.9992	$1.081\ 10^{-8}$	2768	0.9992	0.9992
Landmarks Std	37.41	4.56	2768	37.24	37.58

Confidence intervals are computed at 95% level for each variable based on 2768 observations.
Video Std: Standard deviation of video brightness
PPG Std: Standard deviation of photoplethysmography signals
Landmarks Std: Standard deviation of facial landmarks
CI Low: Lower bound of the 95% confidence interval
CI High: Upper bound of the 95% confidence interval

Next, we sought to determine whether the standard deviations of PPG signals and facial landmarks influenced the variability in video brightness via an interaction effect. We employed a multiple linear regression model incorporating an interaction term, as illustrated in Figure 1. The model yielded a coefficient of $4.104\ 10^8$ for PPG standard deviation and $1.164\ 10^7$ for landmarks standard deviation, neither of which reached statistical significance at the traditional level (p-values of 0.139 and 0.107 respectively). Importantly, the interaction between PPG and landmarks standard deviations produced a coefficient of $-1.165\ 10^7$, with a similarly non-significant p-value of 0.107. These results suggest that while each parameter contributes individually, their interactive effect on video brightness variability is not statistically discernible within this dataset.

Finally, to assess the overall effectiveness of the interaction model, we calculated the R-squared value, a measure of explained variance. The R-squared value was 0.003543, indicating that the interaction model explained very little of the variance in video brightness standard deviation. This underscores the limitation of the current model in capturing the complexity of factors influencing video brightness stability.

In summary, these results reveal distinct individual variabilities within physiological and facial landmark data, but suggest that the interaction of these variabilities does not significantly drive changes in video brightness.

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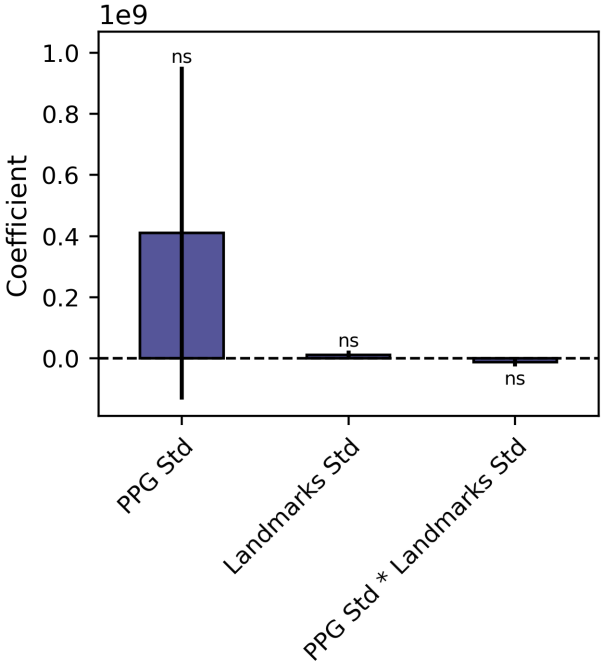

Figure 1: Interaction Effect of PPG and Landmarks on Video Brightness Variation The interaction term represents a multiplicative effect between PPG and landmarks standard deviation coefficients. PPG Std: Standard deviation of photoplethysmography signals. Landmarks Std: Standard deviation of facial landmarks. Coefficient: Estimated coefficient of the model. Standard Error: Standard error of the estimated coefficient. P-value: Probability value indicating significance. Confidence Interval: 95% confidence interval for the coefficient. PPG Std * Landmarks Std: Interaction between PPG and Landmarks standard deviation. Significance: ns $p >= 0.01$, * $p < 0.01$, ** $p < 0.001$, *** $p < 0.0001$.

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The model's low R-squared value further highlights the need for exploration of additional factors that may contribute to video brightness variability.

Discussion

In this study, we explored the interaction between physiological signals and facial landmarks concerning video brightness, an area not extensively covered in previous research [1, 2]. The role of photoplethysmography (PPG) signals for physiological assessment [3], paired with facial landmarks for robust facial recognition [1], creates a complex dynamic influencing video analytics systems. Although numerous studies have examined individual components such as video brightness [12] and physiological signals [13], our research specifically investigated their interaction effects.

We employed a multiple linear regression model to evaluate the joint influence of PPG and facial landmarks on video brightness variability. Descriptive statistics highlighted inherent variability within the dataset, whereas regression analysis underscored that neither individual standard deviations of PPG and landmarks nor their interaction significantly impacted video brightness. Compared to studies demonstrating the enhancement of video quality through signal prediction [4] and perception-based methods using physiological data [6], our findings indicate a limited explanatory power of these interactions for predicting video brightness.

However, this study is not without limitations. The dataset, derived from SCAMPS.csv on the Analemma AI platform, might not encompass the full range of variability present in real-world settings, possibly limiting the generalizability of our findings. Additionally, the low R-squared value of our regression model signifies that other unexamined variables may play a significant role in influencing video brightness. Factors such as environmental lighting conditions and additional physiological indicators, which were not part of this study, might contribute to unexplained variance.

In conclusion, this research provides valuable insights into the interaction effects of PPG signals and facial landmarks on video brightness variability. Although the interaction does not significantly impact brightness, as evidenced by our findings, the study highlights the complexity of factors influencing video analytics. Future research should consider incorporating additional environmental and physiological variables to enhance model robustness and predictive capability. This will be imperative in developing advanced video analytic systems that are resilient to diverse physiological and environmental fluctuations.

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Methods

Data Source

The dataset for this study was obtained from the Analemma AI platform and encompasses a single data file known as SCAMPS.csv. This file contains variables relevant to exploring the relationship between photoplethysmography (PPG) signals, facial landmark deviations, and video characteristics. The dataset offers a comprehensive collection of measurements concerning video brightness variability, variability in physiological signals derived from PPG, and variability in facial landmark measures.

Data Preprocessing

The dataset was imported directly in its available form without necessitating any additional preprocessing steps. All variables necessary for analysis were already formatted numerically. As the core focus lay in analyzing interactions among the original dataset's variables, no transformation was needed.

Data Analysis

The analysis began with a preliminary step of computing descriptive statistics which included the mean, standard deviation, sample size, and 95% confidence intervals for each of the primary variables: video brightness variability, PPG variability, and facial landmark variability. This provided a foundational understanding of the distribution and variability inherent within these measurements.

Following this descriptive assessment, the primary analysis was conducted using a multiple linear regression model. This model was designed to investigate the interaction effect of PPG variability in conjunction with facial landmark variability on the variability of video brightness. The model incorporated an interaction term to specifically gauge the combined influence of the physiological and facial variables relative to the video characteristic.

The statistical output from the regression model included estimates of the coefficients, their associated standard errors, p-values, and confidence intervals, providing insights into the magnitude and significance of the interactions. Additionally, the R-squared value from the model was noted, contributing to the understanding of the model's explanatory power regarding variability in video brightness based on the predictor variables. The

findings were subsequently formatted for interpretation and dissemination within scientific tables.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

References

[1] Xiaoqiang Li, Kaiyuan Wu, and Shaohua Zhang. Landmark-in-facial-component: Towards occlusion-robust facial landmark localization. *Image and Vision Computing*, 151, 2024.

[2] Dasol Jeong, Injae Lee, and Joonki Paik. Audio-to-facial landmarks generator for talking face video synthesis. 2023.

[3] Suresh R. Devasahayam. *Discrete Signal Processing for Physiological Signals*. Springer Singapore, 2019.

[4] Andrei Cernasov. Brightness prediction enhances image quality in video cameras. *SPIE Newsroom*, 2006.

[5] Sik-Ho Tsang, Yui-Lam Chan, and Wan-Chi Siu. Region-based weighted prediction for coding video with local brightness variations. *IEEE Transactions on Circuits and Systems for Video Technology*, 23(3), 2013.

[6] Pablo Arnau-Gonzalez, Turke Althobaiti, Stamos Katsigiannis, and Naeem Ramzan. Perceptual video quality evaluation by means of physiological signals. 2017.

[7] Zitong Yu, Xiaobai Li, and Guoying Zhao. Facial-video-based physiological signal measurement: Recent advances and affective applications. *IEEE Signal Processing Magazine*, 38(6), 2021.

[8] Libin Zhang, Tao Peng, Xiaolu Li, Hanguang Xiao, and Xuewei Chen. Srmformer: A physiological-prior multi-scale periodic attention transformer with raw-video mae for remote physiological measurement. *Signal, Image and Video Processing*, 19(10), 2025.

[9] Mike Fritz and Paul D. Berger. *Can you relate in multiple ways? Multiple linear regression and stepwise regression*. Elsevier, 2015.

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[10] Joseph Antonelli, Giovanni Parmigiani, and Francesca Dominici. High-dimensional confounding adjustment using continuous spike and slab priors. *Bayesian Analysis*, 14(3), 2019.

[11] Divya D. and Bhraguram T.M. *Outlier Detection in Multiple Linear Regression*. IGI Global, 2014.

[12] *Contrast, brightness, contrast, and brightness*. Elsevier, 2012.

[13] Luca Iozzia, Luca Cerina, and Luca Mainardi. Relationships between heart-rate variability and pulse-rate variability obtained from video-ppg signal using zca. *Physiological Measurement*, 37(11), 2016.

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A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0105 from https://analemma.ai/fars/.
    File: SCAMPS.csv

## Data Files
The dataset consists of 1 data file:

### "SCAMPS.csv"
Data file for FA0105. File: SCAMPS.csv. Explore columns to find
    statistically testable hypotheses.
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd
import numpy as np

# Load the dataset
data = pd.read_csv("SCAMPS.csv")

# Open file to write the exploration summary
with open("data_exploration.txt", "w") as f:

    # Data Size
    f.write("# Data Size\n")
    f.write(f"Number of Rows: {data.shape[0]}\n")
    f.write(f"Number of Columns: {data.shape[1]}\n\n")

    # Summary Statistics
    f.write("# Summary Statistics\n")
    summary_stats = data.describe(include='all')
    f.write(summary_stats.to_string())
    f.write("\n\n")

    # Categorical Variables
    f.write("# Categorical Variables\n")
    categorical_columns = data.select_dtypes(include=['object']).
        ↪ columns
    if categorical_columns.empty:
        f.write("# Not Applicable\n")
```

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```
else:
    for col in categorical_columns:
        most_common_value = data[col].mode()[0]
        f.write(f"Column: {col}\n")
        f.write(f"Unique values: {data[col].nunique()}\n")
        f.write(f"Most common value: {most_common_value}\n")
        f.write("\n")
f.write("\n")

# Missing Values
f.write("# Missing Values\n")
missing_counts = data.isnull().sum()
if missing_counts.sum() == 0:
    f.write("# Not Applicable\n")
else:
    for col, count in missing_counts.items():
        if count > 0:
            f.write(f"{col}: {count} missing values\n")
f.write("\n")

# Any other relevant summaries
# Detailing the special numeric values that might stand for
    ↪ unknown/undefined if any found
f.write("# Special Numeric Values\n")
special_numeric_counts = (data == 999).sum()  # Example if 999
    ↪  stands for special value
if any(special_numeric_counts):
    for col, count in special_numeric_counts.items():
        if count > 0:
            f.write(f"Column '{col}': {count} special numeric
                ↪ values (999)\n")
else:
    f.write("# Not Applicable\n")
f.write("\n")

# Add more summary or explore relevant things if needed
# Example: summary for numerical values
numerical_columns = data.select_dtypes(include=[np.number]).
    ↪ columns
f.write("# Numerical Columns Overview\n")
if numerical_columns.empty:
    f.write("# Not Applicable\n")
else:
    for col in numerical_columns:
        f.write(f"Column: {col}, Min: {data[col].min()}, Max:
            ↪ {data[col].max()}\n")
f.write("\n")
```

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B.2 Code Description

The code is designed to perform a comprehensive exploration of the dataset contained in the file "SCAMPS.csv". The analysis is systematically organized into several key steps, each targeting specific aspects of the dataset to provide a clear understanding of its structure and components. The purpose of the data exploration is to summarize and identify characteristics that are essential for further statistical analysis and hypothesis testing.

B.3 Data Size Assessment

The initial phase of the exploration entails a basic examination of the dataset's dimensions, detailing the number of rows and columns it comprises. This information establishes the overall scale of the dataset and provides a preliminary context for the subsequent analysis.

B.4 Summary Statistics Computation

Subsequent to evaluating the data size, the code generates comprehensive summary statistics for all columns using the `describe` function. This function computes metrics such as mean, standard deviation, and quartiles, among others, for both numerical and categorical data. This step provides insights into the distribution and central tendencies of the variables.

B.5 Categorical Variables Analysis

The code identifies and analyzes categorical variables by extracting columns with an object data type. For each categorical column, the number of unique values is determined, and the most common value (mode) is reported. This analysis assists in understanding the variability within categorical data and identifying potential variables of interest for segmentation or further categorical analysis.

B.6 Missing Values Investigation

Handling missing data is crucial for robust analysis. The code highlights columns with missing values by calculating the sum of null entries per column. This information helps in identifying the need for data cleaning or imputation strategies prior to advanced analyses.

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B.7 Special Numeric Values Detection

The code includes a provision for detecting special numeric values that may represent undefined or unknown data points, such as the placeholder ”999”. This step ensures that such values are accounted for and addressed to avoid skewed analysis results.

B.8 Numerical Columns Overview

Finally, the code provides an overview of numerical columns, reporting the minimum and maximum values for each. This step helps in understanding the range and scope of the numerical data, potentially identifying outliers or unusual data points that warrant further investigation.

These steps collectively constitute a thorough exploratory data analysis process, providing descriptive insights and identifying characteristics crucial for hypothesis generation and testing in subsequent research phases.

B.9 Code Output

data_exploration.txt

```
# Data Size
Number of Rows: 2768
Number of Columns: 24

# Summary Statistics
       Unnamed: 0

           file video_dtype         video_shape video_min
           video_max video_mean video_std ppg_dtype ppg_shape
           ppg_min ppg_max ppg_mean   ppg_std landmarks_dtype
           landmarks_shape landmarks_min landmarks_max
           landmarks_mean landmarks_std face_square video_duration
            patient_id  fold
count         2768
                                                                2768
           2768                2768      2768      2768       2768
          2768      2768      2768    2768    2768     2768
    2768            2768            2768          2768
    2768           2768          2768        2768            2768
          2768  2768
unique         NaN
                                                                2768
              1                1612       NaN       NaN         NaN
           NaN          1          1     NaN     NaN      NaN
    NaN                1                1           NaN
```

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```
    NaN            NaN           NaN         NaN            NaN
          2768   NaN
top           NaN  /gim/lv03/datasets/rppg/preprocessed/
    SCAMPS__P000001__0.npz       uint8  (600, 113, 154, 3)
    NaN       NaN        NaN       NaN   float32    (600,)     NaN
         NaN      NaN       NaN           int16    (600, 68, 2)
              NaN           NaN            NaN           NaN
            NaN            NaN    P000001   NaN
freq          NaN
                                                                1
           2768                  10       NaN       NaN        NaN
           NaN      2768      2768     NaN     NaN      NaN
    NaN            2768            2768           NaN
    NaN            NaN           NaN         NaN            NaN
             1   NaN
mean         1384
                                                              NaN
            NaN                 NaN         0     234.2      42.67
         47.41       NaN       NaN       0   3.305    1.081
    0.9992             NaN             NaN             0
    150.7          65.68         37.41    1.85e+04            600
           NaN 1.999
std         799.4
                                                              NaN
            NaN                 NaN         0     36.52      18.27
         16.88       NaN       NaN       0  0.2608   0.1754 1.081e
    -08             NaN             NaN             0
    17.19          9.132          4.56        4214              0
           NaN 1.414
min             0
                                                              NaN
            NaN                 NaN         0        64       6.66
         7.731       NaN       NaN       0   3.042   0.6759
    0.9992             NaN             NaN             0
              102          46.37         24.58        9690
               600        NaN     0
25%         691.8
                                                              NaN
            NaN                 NaN         0       222      29.31
         34.96       NaN       NaN       0    3.14   0.9468
    0.9992             NaN             NaN             0
              145          60.26         33.93       16074
               600        NaN     1
50%          1384
                                                              NaN
            NaN                 NaN         0       255      39.49
         45.48       NaN       NaN       0   3.205    1.082
    0.9992             NaN             NaN             0
              154          64.45            39       17938
```

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```
              600        NaN     2
75%          2075
                                                                NaN
           NaN                 NaN         0       255      52.55
        57.23       NaN       NaN       0   3.341    1.215
   0.9992             NaN             NaN             0
             159          69.06         40.99   2.028e+04
              600        NaN     3
max          2768
                                                                NaN
           NaN                 NaN         0       255      144.4
        116.2       NaN       NaN       0   4.236    1.463
   0.9992             NaN             NaN             0
             310          137.1         51.56       63240
              600        NaN     4

# Categorical Variables
Column: file
Unique values: 2768
Most common value: /gim/lv03/datasets/rppg/preprocessed/
   SCAMPS__P000001__0.npz

Column: video_dtype
Unique values: 1
Most common value: uint8

Column: video_shape
Unique values: 1612
Most common value: (600, 113, 154, 3)

Column: ppg_dtype
Unique values: 1
Most common value: float32

Column: ppg_shape
Unique values: 1
Most common value: (600,)

Column: landmarks_dtype
Unique values: 1
Most common value: int16

Column: landmarks_shape
Unique values: 1
Most common value: (600, 68, 2)

Column: patient_id
Unique values: 2768
Most common value: P000001
```

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```
# Missing Values
# Not Applicable

# Special Numeric Values
Column 'Unnamed: 0': 1 special numeric values (999)

# Numerical Columns Overview
Column: Unnamed: 0, Min: 0, Max: 2768
Column: video_min, Min: 0.0, Max: 0.0
Column: video_max, Min: 64.0, Max: 255.0
Column: video_mean, Min: 6.66             , Max: 144.4
Column: video_std, Min: 7.731            , Max: 116.2
Column: ppg_min, Min: 0.0, Max: 0.0
Column: ppg_max, Min: 3.0415816, Max: 4.235629
Column: ppg_mean, Min: 0.67585063, Max: 1.4631913
Column: ppg_std, Min: 0.99916625, Max: 0.99916637
Column: landmarks_min, Min: 0, Max: 0
Column: landmarks_max, Min: 102, Max: 310
Column: landmarks_mean, Min: 46.37            , Max: 137.1
Column: landmarks_std, Min: 24.58            , Max: 51.56
Column: face_square, Min: 9690, Max: 63240
Column: video_duration, Min: 600, Max: 600
Column: fold, Min: 0, Max: 4
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```python
# IMPORT
from my_utils import df_to_latex, df_to_figure
import pickle
import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf

# LOAD DATA
# Load the dataset assuming it's located at the specified
    ↪ directory path.
scamps_data = pd.read_csv('SCAMPS.csv')

# DATASET PREPARATIONS
# No dataset preparations are needed.
```

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```python
# DESCRIPTIVE STATISTICS
## Table df_desc_stat:
caption = "Descriptive statistics of video_std, ppg_std,
    ↪ landmarks_std"
# Calculate relevant descriptive statistics for scientific tables
desc_stats = scamps_data[['video_std', 'ppg_std', 'landmarks_std'
    ↪ ]].agg(['mean', 'std', 'count'])
desc_stats = desc_stats.T  # Transpose for better readability in
    ↪ the table
desc_stats['ci_low'] = desc_stats['mean'] - 1.96 * desc_stats['std
    ↪ '] / np.sqrt(desc_stats['count'])
desc_stats['ci_high'] = desc_stats['mean'] + 1.96 * desc_stats['
    ↪ std'] / np.sqrt(desc_stats['count'])
df_to_latex(desc_stats, 'df_desc_stat', caption=caption)

# PREPROCESSING
# No preprocessing is needed, because variables needed are already
    ↪  in numerical form

# ANALYSIS
## Table df_interaction_effect:
caption = "Interaction effect between ppg_std and landmarks_std on
    ↪  video_std"

# Fit multiple linear regression model including interaction term
model_formula = 'video_std ~ ppg_std * landmarks_std'
model = smf.ols(formula=model_formula, data=scamps_data).fit()

# Prepare data for LaTeX Table
df_interaction_effect = pd.DataFrame({
    'coef': model.params,
    'stderr': model.bse,
    'p_value': model.pvalues,
    'conf_int': list(zip(model.conf_int()[0], model.conf_int()[1])
        ↪ )
})
df_to_latex(df_interaction_effect, 'df_interaction_effect',
    ↪ caption=caption)

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': len(scamps_data),
    'R-squared of interaction model': model.rsquared,
    # Any other additional results can be added here
}

with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

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C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.

     `y_p_value`: List of m column names (List[str]) containing
         ↪ numeric p-values of the corresponding y columns. These
         ↪ numeric values will be automatically converted by
         ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
         ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.
```

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```
Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

C.3 Code Description

The analysis of the SCAMPS dataset was performed using a combination of descriptive statistics and regression analysis to explore the statistical relationships between the standard deviations of video signal, pulse photoplethysmography (PPG), and facial landmarks data.

C.4 Descriptive Statistics

To begin, descriptive statistics were calculated for three specific columns: ‘video_std‘, ‘ppg_std‘, and ‘landmarks_std‘. These metrics represent the variability inherent in video data, PPG data, and facial landmark data, respectively. The primary purpose of this step was to provide a summary of central tendencies and variabilities of these variables, serving as a foundational understanding for further analysis. Specifically, the mean, standard deviation, and count of observations were computed for each variable. Confidence intervals (95%) for the mean were also calculated to provide an estimate range, offering more insight into the precision of the means.

C.5 Multiple Linear Regression Analysis

A multiple linear regression analysis was conducted to examine the influence of PPG and facial landmarks standard deviations on the video standard deviation. The model included an interaction term between ‘ppg_std‘ and

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‘landmarks_std‘ to investigate whether the relationship between PPG variability and video variability differs at different levels of facial landmarks variability. The regression model was specified using the ‘statsmodels‘ library with the formula:

$$\text{video_std} \sim \text{ppg_std} * \text{landmarks_std}$$

where the interaction term assesses whether there is a significant combined effect of PPG and landmarks variabilities on video variability.

C.6 Result Presentation and Storage

The results from the regression analysis, including coefficients, standard errors, p-values, and confidence intervals for each predictor and the interaction term, were formatted into a table for presentation in LaTeX. Furthermore, key model statistics such as the R-squared value and the total number of observations were stored for future reference, emphasizing the model's explanatory power and the scope of the dataset.

Overall, the analytical steps were structured to initially summarize the data distribution and then explore deeper statistical relationships, focusing on potential interaction effects that might offer new insights into the relationships within the data.

C.7 Code Output

df_desc_stat.pkl

```
                mean       std count ci_low ci_high
video_std      47.41     16.88  2768  46.78   48.04
ppg_std       0.9992 1.081e-08  2768 0.9992  0.9992
landmarks_std  37.41      4.56  2768  37.24   37.58
```

df_interaction_effect.pkl

```
                            coef    stderr p_value
                                                   conf_int
Intercept               -4.1e+08 2.774e+08   0.139  (-9.539e+08,
    1.339e+08)
ppg_std                4.104e+08 2.776e+08   0.139   (-1.34e+08,
    9.547e+08)
landmarks_std          1.164e+07 7.219e+06   0.107    (-2.51e+06,
    2.58e+07)
ppg_std:landmarks_std -1.165e+07 7.225e+06   0.107  (-2.582e+07,
    2.513e+06)
```

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additional_results.pkl

```
{
    'Total number of observations': 2768,
    'R-squared of interaction model': 0.003543,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
all_mapping: AbbrToNameDef = {
    # Rename and provide glossary definitions for technical terms:
    'video_std': ('Video Std', 'Standard deviation of video
        ↪ brightness'),
    'ppg_std': ('PPG Std', 'Standard deviation of
        ↪ photoplethysmography signals'),
    'landmarks_std': ('Landmarks Std', 'Standard deviation of
        ↪ facial landmarks'),
    'coef': ('Coefficient', 'Estimated coefficient of the model'),
    'stderr': ('Standard Error', 'Standard error of the estimated
        ↪ coefficient'),
    'p_value': ('P-value', 'Probability value indicating
        ↪ significance'),
    'conf_int': ('Confidence Interval', '95% confidence interval
        ↪ for the coefficient'),
    'ppg_std:landmarks_std': ('PPG Std * Landmarks Std', '
        ↪ Interaction between PPG and Landmarks standard deviation
        ↪ '),
    # Added entries for missing glossary keys
    'ci_low': ('CI Low', 'Lower bound of the 95% confidence
        ↪ interval'),
    'ci_high': ('CI High', 'Upper bound of the 95% confidence
        ↪ interval')
}

# Process df_desc_stat
df_desc_stat = pd.read_pickle('df_desc_stat.pkl')

# Not Applicable under # Format values:
```

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```
# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_desc_stat, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_desc_stat.rename(columns=abbrs_to_names, index=abbrs_to_names,
    ↪ inplace=True)

df_to_latex(
    df_desc_stat, 'df_desc_stat_formatted',
    caption="Descriptive Statistics of Standard Deviations",
    note="Confidence intervals are computed at 95% level for each
        ↪ variable based on 2768 observations.",
    glossary=glossary
)

# Process df_interaction_effect
df_interaction_effect = pd.read_pickle('df_interaction_effect.pkl'
    ↪ )

# Not Applicable under # Format values:

# Remove the Intercept for plotting
df_interaction_effect = df_interaction_effect.drop(index=['
    ↪ Intercept'])

# Rename rows and columns:
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_interaction_effect, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_interaction_effect.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# Prepare coefficient values for plotting
df_to_figure(
    df_interaction_effect, 'df_interaction_effect_formatted',
    caption="Interaction Effect of PPG and Landmarks on Video
        ↪ Brightness Variation",
    note="The interaction term represents a multiplicative effect
        ↪ between PPG and landmarks standard deviation
        ↪ coefficients.",
    glossary=glossary,
    kind='bar',  # Explicitly specify the kind of plot
    y=['Coefficient'],
    y_ci=['Confidence Interval'],
    y_p_value=['P-value'],
)
```

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D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
    ):
    """
    Save a DataFrame `df` and create a LaTeX figure.
    Parameters, for LaTex embedding of the figure:
    `filename`: Filename for the figure.
    `caption`: Caption for the figure.
    `note`: Note to be added below the figure caption.
    `glossary`: Glossary for the figure.

    Parameters for df.plot():
    `x`: Column name for x-axis (index by default).
    `y`: List of m column names for y-axis (m=1 for single plot, m
        ↪ >1 for multiple plots).
    `kind`: only bar is allowed.
    `logx` / `logy` (bool): log scale for x/y axis.
    `xlabel`: Label for the x-axis.
    `ylabel`: Label for the y-axis.

    `y_ci`: Confidence intervals for errorbars.
        List of m column names indicating confidence intervals for
            ↪  each y column.
        Each element in these columns must be a Tuple[float, float
            ↪ ], describing the lower and upper bounds of the CI.
```

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```
    `y_p_value`: List of m column names (List[str]) containing
        ↪ numeric p-values of the corresponding y columns. These
        ↪ numeric values will be automatically converted by
        ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
        ↪ plotted above the error bars.

    If provided, the length of `y_ci`, and `y_p_value` should be
        ↪ the same as of `y`.

    Example:
    Suppose, we have:

    df_lin_reg_longevity = pd.DataFrame({
        'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
            ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
        'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
            ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
            ↪ [0.0001, ...],
    }, index=['var1', ...])

    then:
    df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
        ↪ caption='Coefficients of ...', kind='bar',
        y=['adjusted_coef', 'unadjusted_coef'],
        y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
        y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
    """


def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

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D.3 Code Output

df_desc_stat_formatted.pkl

```
\begin{table}[h]
\caption{Descriptive Statistics of Standard Deviations}
\label{table:df-desc-stat-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrrrrr}
\toprule
 & mean & std & count & CI Low & CI High \\
\midrule
\textbf{Video Std} & 47.41 & 16.88 & 2768 & 46.78 & 48.04 \\
\textbf{PPG Std} & 0.9992 & 1.081e-08 & 2768 & 0.9992 & 0.9992 \\
\textbf{Landmarks Std} & 37.41 & 4.56 & 2768 & 37.24 & 37.58 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item Confidence intervals are computed at 95\% level for each
    variable based on 2768 observations.
\item \textbf{Video Std}: Standard deviation of video brightness
\item \textbf{PPG Std}: Standard deviation of photoplethysmography
     signals
\item \textbf{Landmarks Std}: Standard deviation of facial
    landmarks
\item \textbf{CI Low}: Lower bound of the 95\% confidence interval
\item \textbf{CI High}: Upper bound of the 95\% confidence
    interval
\end{tablenotes}
\end{threeparttable}
\end{table}
```

df_interaction_effect_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_interaction_effect_formatted.png}
\caption{Interaction Effect of PPG and Landmarks on Video
    Brightness Variation
The interaction term represents a multiplicative effect between
    PPG and landmarks standard deviation coefficients.
PPG Std: Standard deviation of photoplethysmography signals.
Landmarks Std: Standard deviation of facial landmarks.
Coefficient: Estimated coefficient of the model.
Standard Error: Standard error of the estimated coefficient.
P-value: Probability value indicating significance.
```

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```
Confidence Interval: 95\% confidence interval for the coefficient.
PPG Std * Landmarks Std: Interaction between PPG and Landmarks
    standard deviation.
Significance: ns p $>$= 0.01, * p $<$ 0.01, ** p $<$ 0.001, *** p
    $<$ 0.0001.}
\label{figure:df-interaction-effect-formatted}
\end{figure}
% This latex figure presents "df_interaction_effect_formatted.png
    ",
% which was created from the df:
%
% index,"Coefficient","Standard Error","P-value","Confidence
    Interval"
% "PPG Std",4.104e+08,2.776e+08,0.139,(-1.34e+08, 9.547e+08)
% "Landmarks Std",1.164e+07,7.219e+06,0.107,(-2.51e+06, 2.58e+07)
% "PPG Std * Landmarks Std",-1.165e+07,7.225e+06,0.107,(-2.582e
    +07, 2.513e+06)
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', y=['Coefficient'])
%
% Confidence intervals for y-values were then plotted based on
    column: ['Confidence Interval'].
%
% P-values for y-values were taken from column: ['P-value'].
%
% These p-values were presented above the data points as stars (
    with significance threshold values indicated in the figure
    caption).
```